GASTROENTEROLOGY · RNT MEDICAL COLLEGE, UDAIPUR · PILOT WORK, NOVEMBER 2026

Can computer-learned models predict survival in liver disease better than a clinical score?

We tested five models on 312 patients with primary biliary cholangitis (PBC). None clearly beat a well-built clinical score. That result tells us exactly what our own cirrhosis study at RNT needs to do.

Prepared for clinical colleagues. No prior knowledge of artificial intelligence (AI) is assumed. Educational summary, not a clinical tool.

0.84

The clinical score was already very good. In 84 of every 100 patient pairs, it correctly ranked who would die first.

+0.016

The best AI-style model added almost nothing: about 1–2 more correctly ranked pairs per 100. Two models did slightly worse.

Same decisions

Used for real decisions (for example, who needs closer follow-up), all models would select patients almost identically.

Our study: yes, modified

RNT should first test existing scores on Indian patients, collect richer data prospectively, and only then build a new model.

WHY WE DID THIS

A rehearsal before our own study

We are planning a prospective study on cirrhosis and portal hypertension at RNT, which may use AI to predict complications and death. Before collecting a single patient's data, we wanted to know two things:

- Can we build and test prediction models **correctly and honestly**, to the reporting standard journals now expect (TRIPOD+AI, BMJ 2024)?
- On real liver data, does AI actually **beat the clinical scores** we already use?

We used a famous, freely available dataset so that nothing about our own patients was involved.

THE BASIC IDEA

You already use prediction models

MELD, Child–Turcotte–Pugh and SOFA are prediction models: they combine a few measurements into one number that estimates risk. A **"machine learning" or AI model** does the same job, but the computer works out the combination from past patients instead of a committee of experts choosing it.

The benchmark in this pilot was the **Mayo risk score for PBC** (Dickson et al., *Hepatology* 1989). It uses five bedside items: age, bilirubin, albumin, prothrombin time and oedema. Try it below.

Mayo PBC risk score (educational)

Printed with the low-risk example patient from the 1989 paper. The online version lets you enter any values.

Age (years) 52

Bilirubin (mg/dL) 0.5

Albumin (g/dL) 4.5

Prothrombin time (seconds) 10.1

Oedema None, no diuretics

Risk score R = **3.12** (below the average of the original 418 patients, 5.07)



Formula: R = 0.871·ln(bilirubin) − 2.53·ln(albumin) + 0.039·age + 2.38·ln(PT) + 0.859·oedema; survival $S(t) = S_0(t)^{\exp(R - 5.07)}$, with $S_0(t)$ published for years 1–7. Note that it uses prothrombin time in seconds, not INR. It was built for PBC in the 1970s–80s and is shown only to explain what a prediction model does.

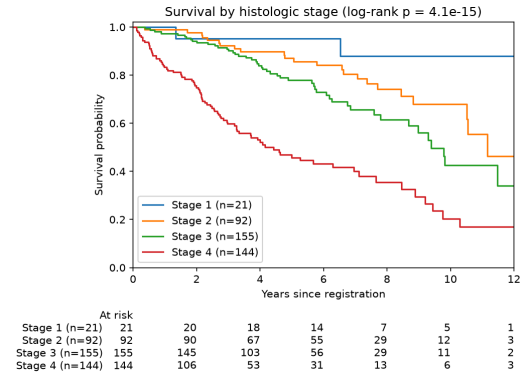
THE PATIENTS

312 patients from a Mayo Clinic trial, 1974–1984

These patients took part in a randomised trial of D-penicillamine versus placebo for PBC, a chronic cholestatic liver disease that progresses to cirrhosis.

- Median age 49.8 years; 88.5% women.
- Followed for a median of 5 years; 125 died and 19 received a liver transplant.
- Five-year survival was 71% overall, and fell from 100% in histological stage 1 to 48% in stage 4.
- For every patient we have baseline age, sex, bilirubin, albumin, prothrombin time, cholesterol, copper, ALP, AST, triglycerides, platelets, ascites, hepatomegaly, spiders, oedema and histological stage.

The data are public (UCI Machine Learning Repository, CC BY 4.0), so this pilot needed no patient consent or ethics approval.



Survival by histological stage, all 418 Mayo patients (Kaplan–Meier). The curves behave as a hepatologist would expect, which is a first check that the data are sound.

WHAT WE DID

Five steps, in this order

1 **Rebuilt the Mayo score and checked it against the 1989 paper.**

We reproduced the paper's worked examples and its published coefficients to three decimal places. This proved our data and code were correct before we tested anything new.

2 **Built five competing models.**

The Mayo variables refitted on these patients (the fair clinical benchmark); a standard Cox model with all 16 variables; and three AI-style models: an “elastic-net” regression that automatically shrinks unhelpful variables, a *random survival forest* (hundreds of decision trees voting), and *gradient boosting* (trees built one after another, each correcting the last).

3 **Tested every model on patients it had never seen.**

Like an exam with unseen questions: each model learned from four-fifths of the patients and was graded on the remaining fifth. We rotated this five times and repeated the whole exercise ten times with different splits, so every patient was a “new” patient many times over (repeated cross-validation).

4 **Graded each model on three clinical questions** (next section), with 95% confidence intervals.

5 **Checked that the answer holds up.** We repeated everything with all 418 patients including non-trial ones, with transplant counted as an outcome, and with transplant treated as a competing event. The conclusion did not change.

HOW WE JUDGED THE MODELS

Three questions a clinician would ask

Discrimination · C-index

Does it put patients in the right order?

Take two patients where one died earlier. Did the model give that patient the higher risk? Count over all such pairs. **0.5 = coin toss, 1.0 = perfect.** Above 0.8 is considered strong.

Calibration · slope, O/E

Are its percentages true?

When the model says “30% risk of death in 5 years”, do about 30 of 100 such patients die? A model can rank well yet over- or under-state risk.

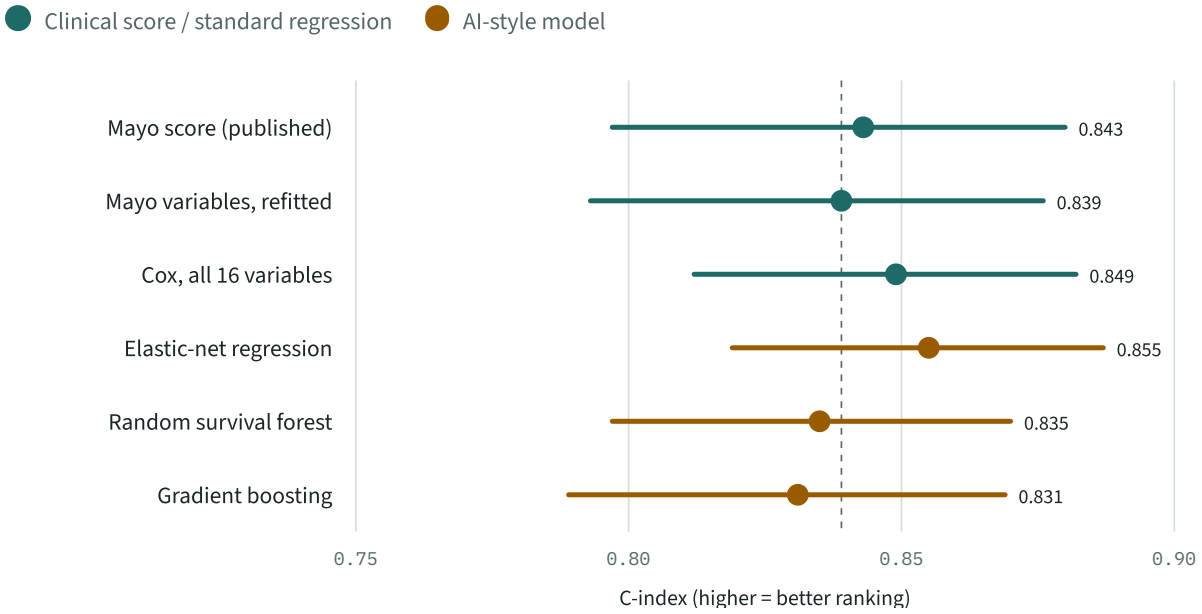
Clinical usefulness · decision curve

Would using it improve decisions?

If we acted on the model (for example, earlier transplant work-up above a chosen risk), would we help more patients than by treating everyone or no one?

RESULTS

The clinical score held its ground



C-index (ranking ability) on unseen patients, with 95% confidence intervals. The dashed line marks the refitted Mayo-variable model. The published Mayo score is slightly flattered because it was originally built on these same patients.

Model	C-index	Difference vs Mayo variables	Are its percentages right?
Mayo variables, refitted (benchmark)	0.839	—	Accurate
Mayo score as published	0.843	+0.005	Accurate
Cox model, all 16 variables	0.849	+0.010 (−0.003 to 0.023)	Slightly too extreme
Elastic-net regression	0.855	+0.016 (0.002 to 0.031)	Overstates risk by about 9%
Random survival forest	0.835	−0.003 (−0.020 to 0.014)	Predictions bunched toward the middle
Gradient boosting	0.831	−0.008 (−0.022 to 0.007)	Overstates risk by about 11%

In plain words

- Every model ranked patients well (C-index about 0.83–0.86).
- Only the elastic-net model was measurably better than the Mayo variables, by about **1–2 extra correct pairs per 100**. That is too small to change care, and it disappeared at 10 years.

What the models learned

When we asked the models which measurements mattered most, they named **bilirubin first by a wide margin**, followed by age, copper, albumin and prothrombin time. Stage, AST and oedema also contributed.

- The two tree-based models, the most “AI-like”, were **no better** and gave less accurate percentages.
- In the decision-curve analysis at 5 years, all six models gave nearly the same benefit at every threshold from 10% to 40%. All were far better than treating everyone alike.

This matches clinical knowledge. The computer found nothing hidden. It rediscovered what hepatologists already knew in 1989.

WHAT THIS MEANS

Four lessons for our own study

1 **AI is not automatically better.**

With a few hundred patients and a handful of good laboratory values, a well-chosen clinical score captures most of the information. A systematic review of 71 clinical studies found the same (Christodoulou et al., *J Clin Epidemiol* 2019).

2 **The data matter more than the algorithm.**

AI has a chance to help when there are many patients, many kinds of information (imaging, elastography, repeated measurements) and complex interactions. This dataset had none of these.

3 **Sample size must be planned.**

By standard formulas (Riley et al., BMJ 2020), 312 patients is only just enough for a 16-variable model, and only if we assume a strong signal. Under more cautious assumptions, about 1,100 patients would be needed.

4 **A score only counts where it has been tested.**

The Mayo score worked here because it was built on these very patients. Whether MELD, MELD 3.0 or any new model works for *our* patients can only be learned by testing it on them.

RECOMMENDATION

Should RNT run a similar study? Yes, in a modified form

Why it is worth doing here

- **Our patients differ.** Most risk scores (MELD, MELD 3.0, CLIF-C) were built on Western patients. Our patients differ in cause of cirrhosis (alcohol, hepatitis B, metabolic liver disease), age at presentation, infections, nutrition and access to care. Acute-on-chronic liver failure is also common in Indian practice.

How it should be modified

1 **First, validate existing scores**

(MELD-Na, MELD 3.0, Child–Pugh, CLIF-C) in a prospective RNT cohort. Standard guidance asks for at least 100, preferably 200, deaths or events, which is feasible.

- **No local validation exists** for how well these scores predict death, decompensation or kidney injury in patients from Rajasthan.
- **We have information the old datasets lack:** FibroScan liver and spleen stiffness, endoscopic variceal grading, and follow-up under Baveno VII and ICA-AKI definitions.
- **The method is now ready.** The validated pipeline from this pilot can be pointed at RNT data with minimal change.

2 **Collect rich, standardised data**

on a common case form: labs, elastography, endoscopy, complications, with fixed follow-up at 3, 6 and 12 months.

3 **Then update or build a model**

only if existing scores fall short. For example, a 1-year mortality model with 10 candidate predictors needs about 700–850 patients followed for a year. This is an illustrative estimate assuming 20–30% one-year mortality and cautious assumptions about model strength; it may require several years or partner centres.

4 **Test the ICU model**

we are developing on large US intensive-care databases (MIMIC-IV, eICU) on RNT's ICU patients with cirrhosis.

> **What we will need from clinical colleagues**
>
> Agreement on outcome definitions; consistent recording on a shared form; reliable follow-up calls; and help with the ethics committee submission. The analysis side is already built and tested.

LIMITS OF THIS PILOT

What this pilot does not show

- PBC is a specific disease of mostly middle-aged women, and these data are 40 years old. The numbers do not apply directly to our cirrhosis patients. The *method and the lesson* do.
- All testing was internal, within one dataset. The true test of any model is a different hospital and population.
- Liver transplant can hide deaths that would have happened. We checked this in three ways and the conclusion held.

GLOSSARY

Terms used above

Machine learning / AI model

A method where the computer finds the best way to combine measurements by studying past patients, instead of an expert choosing the weights.

Cross-validation

Splitting patients into groups, training on some and testing on the rest, then rotating. It shows how a model performs on patients it has not seen, which is what matters at the bedside.

C-index (concordance)

The proportion of patient pairs a model ranks in the correct order of who has the event first. 0.5 is chance; 1.0 is perfect.

Calibration

Agreement between predicted and observed risk. “Too extreme” means high risks are overstated and low risks understated; “too cautious” means the opposite.

Decision-curve analysis

A way to ask whether acting on a model's predictions would do more good than harm compared with treating everyone or no one, across a range of risk thresholds a clinician might choose.

Censoring

When a patient's follow-up ends before the outcome happens (end of study, lost to follow-up, or transplant). Survival methods use the time each patient was observed without assuming they died or survived.

Kaplan–Meier curve

The standard survival curve, which accounts for censored patients.

Confidence interval (95%)

The range in which the true value probably lies. When the interval for a difference includes zero, the difference may be chance.

Data: Mayo Clinic PBC trial, UCI Machine Learning Repository dataset 878 (CC BY 4.0). Benchmark: Dickson ER et al. *Hepatology* 1989;10:1–7. Reporting standard: Collins GS et al. TRIPOD+AI, *BMJ* 2024;385:e078378. Sample size: Riley RD et al. *BMJ* 2020;368:m441.

Full methods, code and the technical report: github.com/atultiwari/cirrhosis-ml-poc. Dept of Gastroenterology, RNT Medical College, Udaipur. Research proof of concept; not for clinical decisions.